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Caloocan City, Metro Manila

RAFAEL G. MADRIGAL

Professional Experience

MANILA WATER COMPANY, INC

Quezon City, Metro Manila

May 2018 to Apr 2020

RESEARCH & PLANNING TECHNICAL MANAGER | Manila Water Philippine Ventures, Inc.

- Facilitated the **Annual Strategy and Execution Planning exercises** of 21 Manila Water Philippine Ventures, Inc. (MWPV) and 3 Manila Water Asia Pacific Pte. Ltd. (MWAP) subsidiaries ahead of schedule
- **Developed and monitored key performance indicators** with middle and senior management and identified and prioritized strategic initiatives for MWAP Group
- Facilitated **Risk Management discussions and reviews** for MWPV and MWAP in preparation for bi-annual Board Risk Oversight Committee reviews.
- Assessed impact of **market trends, regulations, and competition movement** in the waterscape to advise the MWC COO – New Business on investment decisions

Aug 2017 to May 2018

TERRITORY MANAGER | Manila Water Philippine Ventures, Inc.

- Led **business plan development** and takeover of water and wastewater operations of Cloverleaf Balintawak, Vertis North, and Circuit Makati resulting in a **negotiated and secured 5-year water tariffs** which exceeded targets by 15-50%
- Ensured compliance with O&M Agreement and maintained strong client ties with Bonifacio Water Corp (the Client) as **Key Accounts Manager**
- Led a small team of 5 operators and documented and streamlined business and operational processes aligned with ISO 9001:2015 standards.

Dec 2016 to Aug 2017

OIC - TERRITORY MANAGER | Manila Water Philippine Ventures, Inc.

- **Developed relational data model for analysis using PowerPivot** for monthly Billed Volume and Supply Data Establishing data integrity and improving data extraction efficiency vital to success of SAP-ISU project of the business unit
- Managed water operations in Bonifacio Global City resulting in **20% outperformance in Revenue** targets.
- **Increased visibility of brand across all Ayala Land Subdivisions** under operation by standardizing layout and contents of application forms, statements of account, and other communication materials.

Apr 2016 to Dec 2016

MANAGEMENT TRAINEE | Manila Water Philippine Ventures, Inc.

- Led a group that earned the **highest marks during the training's Business Area Immersion**
- **Mapped business processes for** Estate Water by learning from immersions in other subsidiaries of MWPV
- Profiled locators of Bonifacio Global City crucial for data analysis and demand forecasting

Apr 2014 to May 2014

WASTEWATER AGSIP INTERN | Manila Water Company, Inc.

- Analyzed process flow and billed volume data of nine (9) Sewage Treatment Plants (STPs) serving the areas of Cainta, Rizal and Manggahan, Pasig
- Exposed to various wastewater treatment processes implemented and quality assurance tests conducted by the company in all of its treatment plants in the National Capital Region
- Attended various leadership and personal development seminars, under the 6-week Ayala Group Summer Internship Program (AGSIP)

Projects Undertaken

Sukat Online: Virtually Trying on Clothes using Generative Neural Networks

- Applied U-Net Generative Network in style transfer of shirt design to several stock photos

Pandemic Pulse: Topic Evolution of Global COVID-19 Related News Article in 2020

- Modelled global news topics related to COVID-19 in 2020 from the Global Database of Events, Language, and Tone using an Elastic MapReduce cluster on Amazon Web Services, and Apache Spark

DeepRec: Deep Learning Based Hybrid Recommender System

- Developed a Hybrid Recommender System for the movielens data set using PyTorch

Emotion Distribution Recognition of News Articles using Crowd-Annotated Mood Metrics

- Emotion estimation of News Articles using sci-kit learn and gensim

Uncovering a Populist Government: Identifying PH Government's Response to Covid-19 using Cluster Analysis of Pres. Duterte's weekly TV Briefing

- Applied Natural Language Processing and Data Mining techniques to model topics from Pres. Duterte's weekly briefing on COVID 19

Education and Fellowships

March 2020 to Present

MS IN DATA SCIENCE

Asian Institute of Management

- 3x **Dean's Lister** for having Term CGPA 4.51/ 5.00
- **Courses**: Advanced Data Mining, Image Processing, Natural Language Processing, Deep Learning, Machine Learning, Data Visualization, Big Data, Cloud Computing

2010 to 2015

BS CHEMICAL ENGINEERING *(cum laude)*

University of the Philippines Diliman

- **General Weighted Average of 1.508 (GPA = 3.25/4.00);**
- Ranked **74th (Top 10%) out of 720 Engineering graduates;**
- Ranked **14th (Top 16%) out of 87 BSc Chemical Engineering Graduates**

2006 to 2010

HIGH SCHOOL

La Consolacion College Caloocan City

- **Class Salutatorian**
- **Best in Mathematics**
- Recipient, **Mercury Drug Excellence in Mathematics Award**

Sep 2020 to Oct 2020

BLOCKCHAIN BUSINESS SPECIALIST CERTIFICATION PROGRAM

UnionBank of the Philippines

Apr 2018 to Jul 2019

RENEWABLE ENERGY BOOTCAMP (REBOOT)

Friedrich Ebert Stiftung Philippines

Mar 2018 to Jan 2019

FUTURE BRIDGING LEADERSHIP PROGRAM

Asian Institute of Management

Academic Research Work

2019

- **Participatory WASH Assessment in an Indigenous Community in the Philippines** – Two Minute Thesis and Poster Presentation at the 9th International Young Water Professionals Conference in Toronto, Canada

2015 and Older

- **Production of Propylene Glycol via the Hydrogenolysis of Glycerol over Cu-ZnO-Al2O3 Catalysts** – Plant Design Requirement for BSc Chemical Engineering
- **Modelling of Radiation Distribution in a Compound Parabolic Collector (CPC)-based TiO2 Slurry Solar Photocatalytic Reactor** – Research Requirement for BSc Chemical Engineering

Certifications & Licenses

Dec 2020

Natural Language Processing Specialization

Coursera

Nov 2020

Certified Innovation Associate (Level 1)

Global Innovation Management Institute

Sep 2020

Executive Data Science Specialization

Coursera

Nov 2015 to Apr 2022

CHEMICAL ENGINEER

Professional Regulation Commission

Awards and Citations

2019

- President's Pride Due to Performance, Manila Water Philippine Ventures, Inc. [2019]

2015 and Older

- Elected Member (Baccalaureate Level), International Honor Society of Phi Kappa Phi [2015]
- Top Ten of Academic Batch, Department of Chemical Engineering, University of the Philippines Diliman [2014 –2015]
- College Scholar (Dean's Lister), College of Engineering, University of the Philippines Diliman [Mar 2013 –Jun 2015]
- Recipient, Merck, Inc. Scholarship Grant [Nov 2012 -Jun 2015]
- Recipient, Philippine Association of Metropolitan Washington Engineers (PAMWE) Foundation, Inc. Scholarship Grant [Jun 2012–Jan 2014]
- University Scholar (Dean's Lister), College of Engineering, University of the Philippines Diliman [Oct 2010 –Mar 2013]
- Best Engineering Week Day Event Head, UP Chemical Engineering Society, Inc. (UP KEM) [2012]
- Finalist, Noblesse Oblige, a project conceptualization event for student, Engineering Overall Excellence Awards (EOEA) –Individual Category, UP Alliance for Responsive Involvement and Student Empowerment (ARISE) [Jan 2012]
- Consistent High School Honor Student, La Consolacion College - Caloocan [2006 -2010]
- Cadet Captain Officer, 137th Squadron of the Aerospace Cadets of the Philippines [2010]

Seminars and Trainings

2019

- Presenter. Youth Climate Energy Forum. *Friedrich-Ebert-Stiftung Foundation Philippines.* [July 2019]
- Delegate and Poster Presenter. 9th International Young Water Professionals Conference. *International Water Association* [June 2019]
- Trainee. Establishing and Implementing an Integrated Management System (IMS22). *BSI Neville Clark.* [Apr 2019]
- Delegate. 10th International Water Association (IWA) Efficient Conference. *International Water Association* [Jan 2019]

2018

- Delegate. First Bridging Leadership (BL) International Colloquium. *AIM TeaM Energy Center (TEC) for Bridging Leadership in collaboration with Synergos and TeaM Energy Foundation.* [Dec 2018]
- Delegate. Asia Water Forum 2018: "Information, Innovation, Technology". *Asian Development Bank* [Oct 2018]
- Delegate. WaterLinks Forum 2018 "Solving Asia's Urban Water Crisis: The Water-Food-Energy-Climate Change Nexus". *Waterlinks* [Sep 2018]
- Delegate. Philwater 2018: 25th Philippine Water Works Association International Conference & Exhibition. *Philippine Water Works Association* [Sep 2018]
- Fellow. Young Water Leaders' Summit @ Singapore International Water Week 2018. *PUB Singapore's National Water Agency and Singapore Ministry of Environment and Water Resources* [July 2018]
- Delegate. Water Environment Association of the Philippines 13th National Annual Convention "Forging Partnerships in Environmental Protection". *Water Environmental Association of the Philippines.* [Jun 2018]
- Delegate. Asian Forum on Enterprise for Society (AFES) 2018. *Asian Institute of Management Ramon V. Del Rosario (AIM RVR) Center for Corporate Social Responsibility* [Apr 2018]
- Project Innovations Laboratory (InLab): Product Innovation and Development & Business Modelling 101. *Manila Water Total Solutions Corp.* [Sep 2017 –Jan 2018]

2017

- Trainee. Continuing Environmental Education Program (CEEP) for Pollution Control Officers. *Laguna Lake Development Authority (LLDA)* [Dec 2017]
- Delegate. The Filipino Youth Beyond Paris: Acting on Climate, from Paris to Kigali and Beyond. *The Climate Reality Project Philippines* [Nov 2017]

- Trainee. Basic Training Course for Pollution Control Officers (40 hours). *Department of Environment and Natural Resources –Environmental Management Bureau (National Capital Region)* [May 2017]
- Delegate. WaterPhilippines 2017 Exposition. *Philippine Waterworks Association (PWWA)* [Mar 2017]

2016

- Delegate. Sustainable Enterprise Summit 2016: People, Planet, Profit. *EnderunExtension* [May 2016]

2015 and Older

- Attendee. Lecture of Dr. Rigoberto Advincula on "Chemical and Polymer Innovation via Nanostructuring, Lightweighting", and "Additive Manufacturing, and Nanostructure and Polymer Materials for the Oil and Gas Industry". *Engineering Research and Development for Technology (ERDT)* [Jun 2015]
- Delegate. ASEAN Undergraduate Chemical Engineering Education Summit (AUChEEdS) [Jul 2014]
- Trainee. Bridging the Generation Gap: Being a Leader in Innovation While Maintaining Customer Centricity. *Ayala Corporation* [May 2014]
- Goodwill Ambassador. JENESYS 2.0 Exchange Program (Batch 11 – Science and Technology). *Japan International Cooperation Center (JICE) and Department of Science and Technology* [May 2014]
- Participant. Mathematics Trainers' Guild (MTG) Advanced Mathematics Training Program [2010]
- Participant. Mathematics Teachers Association Philippines-Department of Education (MTAP-DepEd) Saturday Programs in Mathematics for Regular and Talented Students [2006 -2010]
- Finalist, Annual Metrobank-MTAP-DepEd Team Mathematics Challenge Private Sectoral Division [2007 -2010]

Advocacy Works

IWA Young Water Professionals (Philippine Chapter)

Diliman, Quezon City

Jul 2019 to Present

ASST. DIR. FOR TRAINING AND DEVELOPMENT

- Spearheads #Kwentubig a water advocacy building platform that harnesses the power of storytelling and personal narratives of young water advocates

2030 Youth Force in the Philippines, Inc.

Maginhawa, Quezon City

Oct 2018 to Present

AMBASSADOR FOR SDG 6: CLEAN WATER AND SANITATION

- Develops programs and projects that advocates for Clean Water and Sanitation (United Nation Sustainable Development Goal 6)
- Promotes awareness on UN SDG 6 through multimedia channels

Apr 2019 to Oct 2019

OIC – NATIONAL DIR. FOR EDUCATION AND RESEARCH

UP Chemical Engineering Society, Inc.

College of Engineering, University of the Philippines Diliman, Quezon City

Oct 2011 to Jun 2015

VICE PRESIDENT FOR FINANCE

- Overhauled organization SOPs for improved cash tracking and events documentation.
- Spearheaded a member development program that focused on project management and events handling.
- Oversaw various FRPs and marketing ventures and drafted programs for future implementation aiming to improve the financial and overall stability of the organization.

Position/s Held:

- Finance Head, Induction Ceremony [Apr 2014]
- Finance Co-head, Engineering Week Team [Oct '2013 -Jan 2014]
- Publicity Coordinator, Fundraising Team (UP KEM Online Shop) [Nov 2012 –May 2013]
- Customer Service Representative, Fundraising Team (UP KEM Online Shop) [Nov 2012 – May 2013]
- Assistant Marketing Head, KEM Gives Back [2013]
- Assistant Head, Chemical Engineering Department Seminar [Jun 2012]
- Finance Head, Freshman Block Handling Team [Apr 2012 –Jun 2012]
- Engineering Week Event Head, Engineering Week Team [Dec 2011]
- Finance Staff, Fundraising Team (Garage Sale) [Oct 2011 –Dec 2011]

UP Career Assistance Program for Engineering Students

College of Engineering, University of the Philippines Diliman, Quezon City

Jun 2013 to Mar 2014

PRODUCTIONS TEAM HEAD

- Secured university establishments for publicity posting two months ahead of schedule through effective coordination with different administration offices;
- Served as secondary liaison between the organization and its printing partners and company sponsors;
- Mainly involved in the production of printed materials for the organization's job fair and seminar series.

Position/s Held

- Logistics Team Member, Company Mixer [Aug 2014 –Dec 2014]
- Internals Team Member [2011]

Engineering Registration Committee

College of Engineering, University of the Philippines Diliman, Quezon City

Apr 2012 to Jun 2015

PEACE AND ORDER OFFICER/ CROWD MARSHALL

- Served as liaison between the student body and administration office during the registration period;
- Assisted students in their registration every semester;
- Responsible for keeping peace and order during the registration period

Other Volunteer Activities

2018 to Present

- Panelist. Our City; Our AdvoQC. *Quezon City Government Environmental Protection and Waste Management Department.* [Nov 2019]
- Panelist. Kids' Innovation Camp 1.1. *Philippine Science High School System.* [Nov 2019]
- Breakout Session ("Matatag") Facilitator and Presenter. NEDA Ambisyon Natin 2040 Symposium and Expo. *National Economic Development Authority.* [Oct 2019]
- Delegate. Sustainable Development Goals (SDG) Bayanihan Summit 2018. *2030 Youth Force in the Philippines, Inc.* [Dec 2018]
- Project Lead/ Convenor, #MagicGulay: Enhancing Social Cohesion in Urban Poor Groups through the development of a co-learning urban [Sep 2018 to Present]
- Volunteer. Save San Roque Alliance [Jun 2019 to Present]

2016

- Tree Planter, #1Bataan: A Guinness World Record Attempt, Provincial Government of Bataan [Jun 2016]
- Tree Planter, La Mesa Eco-Park Tree Planting, Manila Water Foundation and Manila Water Sustainable Department [Jun 2016]
- Volunteer, Brigada Esk'wela: CENTEX Elementary School (Annual Classroom Cleaning and Painting), Ayala Foundation Inc. [Jun 2016]

2015 and Older

- Volunteer Tutor, Diliman Learning Resource Center, University of the Philippines Diliman [Oct 2010 – Mar 2011]